rph8-g15q — formula report

489 inline formulas (section omitted; all rendered), 81 display equations, 0 tables, 12 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
rph8-g15q_ EQ0018 (18)	005	0.837 W|+1	`\begin{aligned} \text { SWAP } L_{1} \text { SWAP } & =L_{2} \\ \left\{X \otimes X, L_{1}\right\} & =0 \end{aligned}`	$\begin{aligned} \text { SWAP } L_{1} \text { SWAP } & =L_{2} \\ \left\{X \otimes X, L_{1}\right\} & =0 \end{aligned}$	$\begin{aligned} \mathtt{SWAP}\, L_{1}\, \mathtt{SWAP} & =L_{2}, \\ \{X \otimes X, L_{1}\} & =0 \end{aligned}$
rph8-g15q_ EQ0065 (D1)	019	0.848 W|+0	`\begin{aligned} d\left(x_{2}, x_{1}\right) & =d y\left(f \left(x_{2}\right), f\left(x_{1}\right)\right) \\ & =d y \left(f\left(x_{1}\right), f\left(x_{2}\right)\right) \\ & =d_{\mathcal{X}}\left(x_{1}, x_{2}\right) \end{aligned}`	$\begin{aligned} d\left(x_{2}, x_{1}\right) & =d y\left(f \left(x_{2}\right), f\left(x_{1}\right)\right) \\ & =d y \left(f\left(x_{1}\right), f\left(x_{2}\right)\right) \\ & =d_{\mathcal{X}}\left(x_{1}, x_{2}\right) \end{aligned}$	$\begin{aligned} d(x_{2}, x_{1}) & =d_{\mathcal{Y}}(f(x_{2}), f(x_{1})) \\ & =d_{\mathcal{Y}}(f(x_{1}), f(x_{2})) \\ & =d_{\mathcal{X}}(x_{1}, x_{2}) \end{aligned}$
rph8-g15q_ EQ0061 (B12)	019	0.890 W|+0	`\begin{array}{llll} Z \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Z, & X \otimes Z, & Z \otimes X, \\ Y \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Y, & X \otimes Y, & Y \otimes X, \end{array}`	$\begin{array}{llll} Z \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Z, & X \otimes Z, & Z \otimes X, \\ Y \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Y, & X \otimes Y, & Y \otimes X, \end{array}$	$\begin{array}{llll} Z \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Z, & X \otimes Z, & Z \otimes X, \\ Y \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Y, & X \otimes Y, & Y \otimes X, \end{array}$
rph8-g15q_ EQ0019 (19)	005	0.921 W|+0	`\begin{array}{lclc} Z \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Z, & Y \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Y, \\ X \otimes Z, & Z \otimes X, & X \otimes Y, & Y \otimes X, \end{array}`	$\begin{array}{lclc} Z \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Z, & Y \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Y, \\ X \otimes Z, & Z \otimes X, & X \otimes Y, & Y \otimes X, \end{array}$	$\begin{array}{llll} Z \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Z, & Y \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Y, \\ X \otimes Z, & Z \otimes X, & X \otimes Y, & Y \otimes X, \end{array}$
rph8-g15q_ EQ0049 (A16)	018	0.933 K|+0	`U(x)=e^{-i \mathcal{L}(x)},`	$U(x)=e^{-i \mathcal{L}(x)},$	$U(x)=e^{-i \mathcal{L}(x)},$
rph8-g15q_EQ0007 (7)	005	0.945 C|-12	`\begin{array}{ll} \alpha_{(0,0)}=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right], & \alpha_{(1,0)}= \left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array} \right], \\ \alpha_{(0,1)}=\left[\begin{array}{rr} -1 & 0 \\ 0 & -1 \end{array}\right], \quad \alpha_{(1,1)}= \left[\begin{array}{rr} 0 & -1 \\ -1 & 0 \end{array} \right] . \end{array}`	$\begin{array}{ll} \alpha_{(0,0)}=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right], & \alpha_{(1,0)}= \left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array} \right], \\ \alpha_{(0,1)}=\left[\begin{array}{rr} -1 & 0 \\ 0 & -1 \end{array}\right], \quad \alpha_{(1,1)}= \left[\begin{array}{rr} 0 & -1 \\ -1 & 0 \end{array} \right] . \end{array}$	$\begin{array}{ll} \alpha_{(0,0)}=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right], & \alpha_{(1,0)}= \left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array} \right], \\ \alpha_{(0,1)}=\left[\begin{array}{rr} -1 & 0 \\ 0 & -1 \end{array}\right], & \alpha_{(1,1)}= \left[\begin{array}{rr} 0 & -1 \\ -1 & 0 \end{array} \right]. \end{array}$
rph8-g15q_ EQ0064 (B9)	019	0.951 S|+1	`\begin{aligned} d_{A}\left(x_{1}, x_{2}\right) & =\sqrt{ \left(x_{1}-x_{2}\right)^{T} V^{T} B V\left(x_{1}-x_{2} \right)} \\ & =\sqrt{\left(V x_{1}-V x_{2}\right)^{T} B \left(V x_{1}-V x_{2}\right)} \\ & =d_{B}\left(V x_{1}, V x_{2}\right) \end{aligned}`	$\begin{aligned} d_{A}\left(x_{1}, x_{2}\right) & =\sqrt{ \left(x_{1}-x_{2}\right)^{T} V^{T} B V\left(x_{1}-x_{2} \right)} \\ & =\sqrt{\left(V x_{1}-V x_{2}\right)^{T} B \left(V x_{1}-V x_{2}\right)} \\ & =d_{B}\left(V x_{1}, V x_{2}\right) \end{aligned}$	$\begin{aligned} d_{A}(x_{1}, x_{2}) & =\sqrt{(x_{1}-x_{2})^{T} V^{T} B V(x_{1}-x_{2})} \\ & =\sqrt{(V x_{1}-V x_{2})^{T} B(V x_{1}-V x_{2})} \\ & =d_{B}(V x_{1}, V x_{2}), \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>rph8-g15q_EQ0008 (8)</td><td>005</td><td>■ 0.962
● W|+0</td><td>\begin{array}{ll} V_{(0,0)}=\mathbb{1}_{2} \otimes \mathbb{1}_{2}, & V_{(1,0)}=\operatorname{SWAP}, \\ V_{(0,1)}=X \otimes X, & V_{(1,1)}=\operatorname{SWAP}(X \otimes X), \end{array}</td><td>$\begin{array}{ll} V_{(0,0)}=\mathbb{1}_{2} \otimes \mathbb{1}_{2}, & V_{(1,0)}=\operatorname{SWAP}, \\ V_{(0,1)}=X \otimes X, & V_{(1,1)}=\operatorname{SWAP}(X \otimes X), \end{array}$</td><td>$\begin{array}{ll} V_{(0,0)}=\mathbb{1}_{2} \otimes \mathbb{1}_{2}, & V_{(1,0)}=\mathtt{SWAP}, \\ V_{(0,1)}=X \otimes X, & V_{(1,1)}=\mathtt{SWAP}(X \otimes X), \end{array}$</td></tr>
<tr><td>rph8-g15q_EQ0046 (A13)</td><td>017</td><td>■ 0.985
● S|+1</td><td>\begin{aligned} y & =\left.\frac{d}{d t} \exp (t y) \right|_{t=0}=\left.\frac{d}{d t}(f \circ \gamma) \right|_{t=0} \quad \text { by Eq. (A12) } \\ & = \left.D f\right|_{1_{G}}\left(\left.\frac{d}{d t} \gamma \right|_{t=0}\right) \quad \text { by the chain rule } \\ & =\left.D f\right|_{1_{G}}\left(\left.\frac{d}{d t} \exp (t x)\right|_{t=0}\right) \quad \text { by Eq. (A11) } \\ & =\mathcal{M}(x) \quad \text { by definition of } \mathcal{M} \end{aligned}</td><td>$\begin{aligned} y & =\left.\frac{d}{d t} \exp (t y) \right|_{t=0}=\left.\frac{d}{d t}(f \circ \gamma) \right|_{t=0} \quad \text { by Eq. (A12) } \\ & = \left.D f\right|_{1_{G}}\left(\left.\frac{d}{d t} \gamma \right|_{t=0}\right) \quad \text { by the chain rule } \\ & =\left.D f\right|_{1_{G}}\left(\left.\frac{d}{d t} \exp (t x)\right|_{t=0}\right) \quad \text { by Eq. (A11) } \\ & =\mathcal{M}(x) \quad \text { by definition of } \mathcal{M} \end{aligned}$</td><td>$\begin{aligned} y & = \frac{d}{dt}\exp(ty)\vert_{t=0} = \frac{d}{dt}(f\circ\gamma)\vert_{t=0} \quad \text{by Eq. (A12)} \\ & = Df\vert_{1_G}\left(\frac{d}{dt}\gamma\vert_{t=0}\right) \quad \text{by the chain rule} \\ & = Df\vert_{1_G}\left(\frac{d}{dt}\exp(tx)\vert_{t=0}\right) \quad \text{by Eq. (A11)} \\ & = \mathcal{M}(x) \quad \text{by definition of } \mathcal{M}. \end{aligned}$ (A</td></tr>
<tr><td>rph8-g15q_EQ0067 (D3)</td><td>019</td><td>■ 0.991
● W|+1</td><td>\begin{aligned} d_{\mathcal{X}}\left(x_{1}, x_{2}\right) & \geqslant d_{\mathcal{Y}}\left(f\left(x_{1}\right), f\left(x_{3}\right)\right)+d_{\mathcal{Y}}\left(f \left(x_{3}\right), f\left(x_{2}\right)\right) \\ & =d_{\mathcal{X}}\left(x_{1}, x_{3}\right)+d_{\mathcal{X}} \left(x_{3}, x_{2}\right) \end{aligned}</td><td>$\begin{aligned} d_{\mathcal{X}}\left(x_{1}, x_{2}\right) & \geqslant d_{\mathcal{Y}}\left(f\left(x_{1}\right), f\left(x_{3}\right)\right)+d_{\mathcal{Y}}\left(f \left(x_{3}\right), f\left(x_{2}\right)\right) \\ & =d_{\mathcal{X}}\left(x_{1}, x_{3}\right)+d_{\mathcal{X}} \left(x_{3}, x_{2}\right) \end{aligned}$</td><td>$\begin{aligned} d_{\mathcal{X}}(x_1, x_2) & \geqslant d_{\mathcal{Y}}(f(x_1), f(x_3)) + d_{\mathcal{Y}}(f(x_3), f(x_2)) \\ & = d_{\mathcal{X}}(x_1, x_3) + d_{\mathcal{X}}(x_3, x_2), \end{aligned}$</td></tr>
<tr><td>rph8-g15q_EQ0027 (27)</td><td>010</td><td>■ 0.994
● N|+0</td><td>d_{\mathrm{Tr}}(\rho, \sigma)=\|\rho-\sigma\|_{1} \equiv \operatorname{Tr}\left[\sqrt{(\rho-\sigma)^{\dagger}(\rho-\sigma)}\right] .</td><td>$d_{\mathrm{Tr}}(\rho, \sigma)=\|\rho-\sigma\|_{1} \equiv \operatorname{Tr}\left[\sqrt{(\rho-\sigma)^{\dagger}(\rho-\sigma)}\right] .$</td><td>$d_{\mathrm{Tr}}(\rho, \sigma) = \|\rho - \sigma\|_1 \equiv \mathrm{Tr}\left[\sqrt{(\rho-\sigma)^{\dagger}(\rho-\sigma)}\right].$</td></tr>
<tr><td>rph8-g15q_EQ0054 (B5)</td><td>018</td><td>■ 0.994
● N|+0</td><td>\left|\psi_{0}\right\rangle=\sqrt{p}|+,+\rangle- \sqrt{1-p}|-,-\rangle</td><td>$\left|\psi_{0}\right\rangle=\sqrt{p}|+,+\rangle- \sqrt{1-p}|-,-\rangle$</td><td>$|\psi_0\rangle = \sqrt{p}|+,+\rangle - \sqrt{1-p}|-,-\rangle$</td></tr>
<tr><td>rph8-g15q_EQ0009 (9)</td><td>005</td><td>■ 0.996
● N|+1</td><td>U\left(x_{1}, x_{2}\right)=R_{Z}\left(x_{1}\right) \otimes R_{Z}\left(x_{2}\right),</td><td>$U\left(x_{1}, x_{2}\right)=R_{Z}\left(x_{1}\right) \otimes R_{Z}\left(x_{2}\right),$</td><td>$U(x_1, x_2) = R_Z(x_1) \otimes R_Z(x_2),$</td></tr>
<tr><td>rph8-g15q_EQ0011 (11)</td><td>005</td><td>■ 0.999
● N|+0</td><td>\left|\psi_{0}\right\rangle=\sqrt{p}|+,+\rangle- \sqrt{1-p}|-,-\rangle,</td><td>$\left|\psi_{0}\right\rangle=\sqrt{p}|+,+\rangle- \sqrt{1-p}|-,-\rangle,$</td><td>$|\psi_0\rangle = \sqrt{p}|+,+\rangle - \sqrt{1-p}|-,-\rangle,$</td></tr>
<tr><td>rph8-g15q_EQ0005 (5)</td><td>005</td><td>■ 1.000
● N|+0</td><td>y\left(x_{1}, x_{2}\right)=y\left(x_{2}, x_{1}\right)=y \left(-x_{1},-x_{2}\right),</td><td>$y\left(x_{1}, x_{2}\right)=y\left(x_{2}, x_{1}\right)=y \left(-x_{1},-x_{2}\right),$</td><td>$y(x_1, x_2) = y(x_2, x_1) = y(-x_1, -x_2),$</td></tr>
<tr><td>rph8-g15q_EQ0016 (16)</td><td>005</td><td>■ 1.000
● N|+0</td><td>V_{g} e^{-i \mathcal{L}(x)} V_{g}^{\dagger}=e^{-i \mathcal{L}\left(\alpha_{g}(x)\right)}</td><td>$V_{g} e^{-i \mathcal{L}(x)} V_{g}^{\dagger}=e^{-i \mathcal{L}\left(\alpha_{g}(x)\right)}$</td><td>$V_g e^{-i\mathcal{L}(x)} V_g^{\dagger} = e^{-i\mathcal{L}(\alpha_g(x))}$</td></tr>
<tr><td>rph8-g15q_EQ0023 (23)</td><td>007</td><td>■ 1.000
● N|+0</td><td>x \mapsto \chi(x)=\sum_{n=0}^{2^{d}-1} x_{n}|n\rangle \mapsto|\Phi(x)\rangle=\frac{\chi(x)}{\|\chi(x)\|},</td><td>$x \mapsto \chi(x)=\sum_{n=0}^{2^{d}-1} x_{n}|n\rangle \mapsto|\Phi(x)\rangle=\frac{\chi(x)}{\|\chi(x)\|},$</td><td>$x \mapsto \chi(x) = \sum_{n=0}^{2^d-1} x_n|n\rangle \mapsto |\Phi(x)\rangle = \frac{\chi(x)}{\|\chi(x)\|},$</td></tr>
<tr><td>rph8-g15q_EQ0029 (29)</td><td>010</td><td>■ 1.000
● S|+1</td><td>d_{B}(\rho, \sigma)=\sqrt{2(1-\sqrt{F(\rho, \sigma)})},</td><td>$d_{B}(\rho, \sigma)=\sqrt{2(1-\sqrt{F(\rho, \sigma)})},$</td><td>$d_B(\rho, \sigma) = \sqrt{2\big(1 - \sqrt{F(\rho, \sigma)}\big)},$</td></tr>
<tr><td>rph8-g15q_EQ0030 (30)</td><td>010</td><td>■ 1.000
● N|+1</td><td>F(\rho, \sigma)=(\operatorname{Tr}[\sqrt{\sqrt{\rho} \sigma \sqrt{\rho}}])^{2}</td><td>$F(\rho, \sigma)=(\operatorname{Tr}[\sqrt{\sqrt{\rho} \sigma \sqrt{\rho}}])^{2}$</td><td>$F(\rho, \sigma) = \left(\mathrm{Tr}\left[\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}\right]\right)^2$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>rph8-g15q_EQ0032 (32)</td><td>013</td><td>1.000 N|+0</td><td><code>F\left(\operatorname{id}_{(\mathcal{X}, \alpha)}\right)=\operatorname{id}_{(\mathcal{X}, \alpha)}=\operatorname{id}_{\mathcal{X}}=\operatorname{id}_{F(\mathcal{X}, \alpha)} .</code></td><td>$F\left(\operatorname{id}_{(\mathcal{X}, \alpha)}\right)=\operatorname{id}_{(\mathcal{X}, \alpha)}=\operatorname{id}_{\mathcal{X}}=\operatorname{id}_{F(\mathcal{X}, \alpha)} .$</td><td>$F(\mathrm{id}_{(\mathcal{X},\alpha)}) = \mathrm{id}_{(\mathcal{X},\alpha)} = \mathrm{id}_{\mathcal{X}} = \mathrm{id}_{F(\mathcal{X},\alpha)}.$</td></tr>
<tr><td>rph8-g15q_EQ0001 (1)</td><td>003</td><td>1.000 K|+0</td><td><code>\rho(x)=U(x)\left|\psi_{0}\right\rangle\left\langle\psi_{0}\right| U(x)^{\dagger} .</code></td><td>$\rho(x)=U(x)\left|\psi_{0}\right\rangle\left\langle\psi_{0}\right| U(x)^{\dagger} .$</td><td>$\rho(x) = U(x)|\psi_0\rangle\langle\psi_0|U(x)^\dagger.$</td></tr>
<tr><td>rph8-g15q_EQ0002 (2)</td><td>004</td><td>1.000 N|+0</td><td><code>\operatorname{Ad}_{V_{g}}(\sigma):=V_{g} \sigma V_{g}^{\dagger}</code></td><td>$\operatorname{Ad}_{V_{g}}(\sigma):=V_{g} \sigma V_{g}^{\dagger}$</td><td>$\mathrm{Ad}_{V_g}(\sigma) := V_g \sigma V_g^\dagger$</td></tr>
<tr><td>rph8-g15q_EQ0003 (3)</td><td>004</td><td>1.000 N|+0</td><td><code>\rho\left(\alpha_{g}(x)\right)=V_{g} \rho(x) V_{g}^{\dagger}</code></td><td>$\rho\left(\alpha_{g}(x)\right)=V_{g} \rho(x) V_{g}^{\dagger}$</td><td>$\rho(\alpha_g(x)) = V_g \rho(x) V_g^\dagger$</td></tr>
<tr><td>rph8-g15q_EQ0004 (4)</td><td>004</td><td>1.000 N|+0</td><td><code>f\left(\alpha_{g}(x)\right)=\beta_{g}(f(x))</code></td><td>$f\left(\alpha_{g}(x)\right)=\beta_{g}(f(x))$</td><td>$f(\alpha_g(x)) = \beta_g(f(x))$</td></tr>
<tr><td>rph8-g15q_EQ0006 (6)</td><td>005</td><td>1.000 N|+0</td><td><code>G=\mathbb{Z}_{2} \times \mathbb{Z}_{2}=\{(0,0),(1,0),(0,1),(1,1)\}</code></td><td>$G=\mathbb{Z}_{2} \times \mathbb{Z}_{2}=\{(0,0),(1,0),(0,1),(1,1)\}$</td><td>$G = \mathbb{Z}_2 \times \mathbb{Z}_2 = \{(0,0),(1,0),(0,1),(1,1)\}$</td></tr>
<tr><td>rph8-g15q_EQ0010 (10)</td><td>005</td><td>1.000 N|+1</td><td><code>R_{Z}(\theta)=e^{-i \theta Z / 2}=\left[\begin{array}{cc} e^{-i \theta / 2} & 0 \\ 0 & e^{i \theta / 2} \end{array}\right]</code></td><td>$R_{Z}(\theta)=e^{-i \theta Z / 2}=\left[\begin{array}{cc} e^{-i \theta / 2} & 0 \\ 0 & e^{i \theta / 2} \end{array}\right]$</td><td>$R_Z(\theta) = e^{-i\theta Z/2} = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix},$</td></tr>
<tr><td>rph8-g15q_EQ0012 (12)</td><td>005</td><td>1.000 N|-1</td><td><code>|+\rangle=\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle) \quad \text { and } \quad|-\rangle=\frac{1}{\sqrt{2}}(|0\rangle-|1\rangle)</code></td><td>$|+\rangle=\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle) \quad \text { and } \quad|-\rangle=\frac{1}{\sqrt{2}}(|0\rangle-|1\rangle)$</td><td>$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad \text{and} \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$</td></tr>
<tr><td>rph8-g15q_EQ0013 (13)</td><td>005</td><td>1.000 K|+0</td><td><code>\rho(x)=U\left(x_{1}, x_{2}\right)\left|\psi_{0}\right\rangle\left\langle\psi_{0}\right| U\left(x_{1}, x_{2}\right)^{\dagger} .</code></td><td>$\rho(x)=U\left(x_{1}, x_{2}\right)\left|\psi_{0}\right\rangle\left\langle\psi_{0}\right| U\left(x_{1}, x_{2}\right)^{\dagger} .$</td><td>$\rho(x) = U(x_1, x_2)|\psi_0\rangle\langle\psi_0|U(x_1, x_2)^\dagger.$</td></tr>
<tr><td>rph8-g15q_EQ0014 (14)</td><td>005</td><td>1.000 N|+0</td><td><code>U(x)=e^{-i \mathcal{L}(x)},</code></td><td>$U(x)=e^{-i \mathcal{L}(x)},$</td><td>$U(x) = e^{-i\mathcal{L}(x)},$</td></tr>
<tr><td>rph8-g15q_EQ0015 (15)</td><td>005</td><td>1.000 N|-1</td><td><code>L_{1}:=\mathcal{L}\left(e_{1}\right) \quad \text { and } \quad L_{2}:=\mathcal{L}\left(e_{2}\right),</code></td><td>$L_{1}:=\mathcal{L}\left(e_{1}\right) \quad \text { and } \quad L_{2}:=\mathcal{L}\left(e_{2}\right),$</td><td>$L_1 := \mathcal{L}(e_1) \quad \text{and} \quad L_2 := \mathcal{L}(e_2),$</td></tr>
<tr><td>rph8-g15q_EQ0017 (17)</td><td>005</td><td>1.000 N|+0</td><td><code>V_{g} \mathcal{L}(x) V_{g}^{\dagger}=\mathcal{L}\left(\alpha_{g}(x)\right)</code></td><td>$V_{g} \mathcal{L}(x) V_{g}^{\dagger}=\mathcal{L}\left(\alpha_{g}(x)\right)$</td><td>$V_g \mathcal{L}(x) V_g^\dagger = \mathcal{L}(\alpha_g(x))$</td></tr>
<tr><td>rph8-g15q_EQ0020 (20)</td><td>006</td><td>1.000 N|-2</td><td><code>L_{1}=\frac{Z}{2} \otimes \mathbb{1}_{2} \quad \text { and } \quad L_{2}=\mathbb{1}_{2} \otimes \frac{Z}{2}</code></td><td>$L_{1}=\frac{Z}{2} \otimes \mathbb{1}_{2} \quad \text { and } \quad L_{2}=\mathbb{1}_{2} \otimes \frac{Z}{2}$</td><td>$L_1 = \frac{Z}{2} \otimes \mathbb{1}_2 \quad \text{and} \quad L_2 = \mathbb{1}_2 \otimes \frac{Z}{2}$</td></tr>
<tr><td>rph8-g15q_EQ0021 (21)</td><td>006</td><td>1.000 N|+0</td><td><code>x \mapsto|\Phi(x)\rangle:=\left(\bigotimes_{k=1}^{d} \exp \left(-\frac{i x_{k}}{2} X_{k}\right)\right)|0 \cdots 0\rangle,</code></td><td>$x \mapsto|\Phi(x)\rangle:=\left(\bigotimes_{k=1}^{d} \exp \left(-\frac{i x_{k}}{2} X_{k}\right)\right)|0 \cdots 0\rangle,$</td><td>$x \mapsto |\Phi(x)\rangle := \left(\bigotimes_{k=1}^{d} \exp\left(-\frac{ix_k}{2} X_k\right)\right)|0 \cdots 0\rangle,$</td></tr>
<tr><td>rph8-g15q_EQ0022 (22)</td><td>006</td><td>1.000 K|+0</td><td><code>\rho(x):=|\Phi(x)\rangle\langle\Phi(x)| .</code></td><td>$\rho(x):=|\Phi(x)\rangle\langle\Phi(x)| .$</td><td>$\rho(x) := |\Phi(x)\rangle\langle\Phi(x)|.$</td></tr>
<tr><td>rph8-g15q_EQ0024 (24)</td><td>007</td><td>1.000 N|+0</td><td><code>\left(x_{0}, x_{1}, \ldots, x_{N-1}\right) \mapsto \frac{\left(x_{0}, x_{1}, \ldots, x_{N-1}, 1\right)}{\sqrt{1+\|x\|^{2}}},</code></td><td>$\left(x_{0}, x_{1}, \ldots, x_{N-1}\right) \mapsto \frac{\left(x_{0}, x_{1}, \ldots, x_{N-1}, 1\right)}{\sqrt{1+\|x\|^{2}}},$</td><td>$(x_0, x_1, \ldots, x_{N-1}) \mapsto \frac{(x_0, x_1, \ldots, x_{N-1}, 1)}{\sqrt{1 + \|x\|^2}},$</td></tr>
<tr><td>rph8-g15q_EQ0025 (25)</td><td>007</td><td>1.000 C|-3</td><td><code>\left(x_{0}, \ldots, x_{N-1}\right) \mapsto \frac{\left(e^{x_{0}}, \ldots, e^{x_{N-1}}, 1\right)}{1+\sum_{j=0}^{N-1} e^{x_{j}}},</code></td><td>$\left(x_{0}, \ldots, x_{N-1}\right) \mapsto \frac{\left(e^{x_{0}}, \ldots, e^{x_{N-1}}, 1\right)}{1+\sum_{j=0}^{N-1} e^{x_{j}}},$</td><td>$(x_0, \ldots, x_{N-1}) \mapsto \frac{(e^{x_0}, \ldots, e^{x_{N-1}}, 1)}{1 + \sum_{j=0}^{N-1} e^{x_j}},$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
rph8-g15q_ EQ0026 (26)	009	■ 1.000 ● K|+0	`d_{\mathcal{X}}\left(x_{1}, x_{2}\right):=d_{\mathcal{Y}} \left(f\left(x_{1}\right), f\left(x_{2}\right)\right)`	$d_{\mathcal{X}}\left(x_{1}, x_{2}\right):=d_{\mathcal{Y}} \left(f\left(x_{1}\right), f\left(x_{2}\right)\right)$	$d_{\mathcal{X}}(x_1, x_2) := d_{\mathcal{Y}}(f(x_1), f(x_2))$
rph8-g15q_ EQ0028 (28)	010	■ 1.000 ● K|+0	`d_{\mathrm{HS}}(\rho, \sigma)=\sqrt{\operatorname{Tr} \left[(\rho-\sigma)^{\dagger}(\rho-\sigma)\right]} .`	$d_{\mathrm{HS}}(\rho, \sigma)=\sqrt{\operatorname{Tr} \left[(\rho-\sigma)^{\dagger}(\rho-\sigma)\right]} .$	$d_{\mathrm{HS}}(\rho, \sigma) = \sqrt{\operatorname{Tr}[(\rho-\sigma)^{\dagger}(\rho-\sigma)]}.$
rph8-g15q_ EQ0031 (31)	013	■ 1.000 ● N|+0	`F(g \circ f)=g \circ f=F(g) \circ F(f) .`	$F(g \circ f)=g \circ f=F(g) \circ F(f) .$	$F(g \circ f) = g \circ f = F(g) \circ F(f).$
rph8-g15q_ EQ0033 (33)	014	■ 1.000 ● N|+0	`U(x)=e^{-i \mathcal{L}(x)},`	$U(x)=e^{-i \mathcal{L}(x)},$	$U(x) = e^{-i\mathcal{L}(x)},$
rph8-g15q_ EQ0034 (A1)	016	■ 1.000 ● K|+0	`U(x)=e^{-i \mathcal{L}(x)},`	$U(x)=e^{-i \mathcal{L}(x)},$	$U(x) = e^{-i\mathcal{L}(x)},$
rph8-g15q_ EQ0035 (A2)	016	■ 1.000 ● N|+0	`\gamma(t)=\exp (t M)`	$\gamma(t)=\exp (t M)$	$\gamma(t) = \exp(tM)$
rph8-g15q_ EQ0036 (A3)	016	■ 1.000 ● K|+0	`f(\exp (W))=\exp (\mathcal{M}(W))`	$f(\exp (W))=\exp (\mathcal{M}(W))$	$f(\exp(W)) = \exp(\mathcal{M}(W))$
rph8-g15q_ EQ0037 (A4)	016	■ 1.000 ● N|+0	`\left[\mathcal{M}\left(W_{1}\right), \mathcal{M} \left(W_{2}\right)\right]=\mathcal{M}\left(\left[W_{1}, W_{2}\right]\right)`	$\left[\mathcal{M}\left(W_{1}\right), \mathcal{M} \left(W_{2}\right)\right]=\mathcal{M}\left(\left[W_{1}, W_{2}\right]\right)$	$[\mathcal{M}(W_1), \mathcal{M}(W_2)] = \mathcal{M}([W_1, W_2])$
rph8-g15q_ EQ0038 (A5)	017	■ 1.000 ● K|+0	`U((s, t))=e^{-i s X-i t Y}`	$U((s, t))=e^{-i s X-i t Y}$	$U((s,t)) = e^{-isX-itY}$
rph8-g15q_ EQ0039 (A6)	017	■ 1.000 ● N|+0	`\begin{aligned} U((s, 0)) U((0, t)) & =e^{-i s X} e^{-i t Y} \\ & \neq e^{-i(s X+t Y)} \\ & =U((s, 0)+(0, t)) \end{aligned}`	$\begin{aligned} U((s, 0)) U((0, t)) & =e^{-i s X} e^{-i t Y} \\ & \neq e^{-i(s X+t Y)} \\ & =U((s, 0)+(0, t)) \end{aligned}$	$\begin{aligned} U((s,0))U((0,t)) &= e^{-isX}e^{-itY} \\ &\neq e^{-i(sX+tY)} \\ &= U((s,0)+(0,t)) \end{aligned}$
rph8-g15q_ EQ0040 (A7)	017	■ 1.000 ● N|+1	`\mathbb{R}_{\alpha}=\left\{r \alpha: \alpha \in \mathbb{R}^{2} \backslash\{0\}, r \in \mathbb{R}\right\},`	$\mathbb{R}_{\alpha}=\left\{r \alpha: \alpha \in \mathbb{R}^{2} \backslash\{0\}, r \in \mathbb{R}\right\},$	$\mathbb{R}_{\alpha} = \{r\alpha \ : \ \alpha \in \mathbb{R}^2 \setminus \{0\},\ r \in \mathbb{R}\},$
rph8-g15q_ EQ0041 (A8)	017	■ 1.000 ● S|+1	`\begin{aligned} U(r \alpha) U(u \alpha) & =e^{-i r W} e^{-i u W} \\ & =e^{-i(r+u) W} \\ & =U(r \alpha+u \alpha), \end{aligned}`	$\begin{aligned} U(r \alpha) U(u \alpha) & =e^{-i r W} e^{-i u W} \\ & =e^{-i(r+u) W} \\ & =U(r \alpha+u \alpha), \end{aligned}$	$\begin{aligned} U(r\alpha)U(u\alpha) &= e^{-irW}e^{-iuW} \\ &= e^{-i(r+u)W} \\ &= U(r\alpha+u\alpha), \end{aligned}$
rph8-g15q_ EQ0042 (A9)	017	■ 1.000 ● K|+0	`\mathbb{R} \xrightarrow{\alpha} \mathbb{R}^{2} \xrightarrow{U} \mathscr{U}\left(\mathbb{C}^{2}\right)`	$\mathbb{R} \xrightarrow{\alpha} \mathbb{R}^{2} \xrightarrow{U} \mathscr{U}\left(\mathbb{C}^{2}\right)$	$\mathbb{R} \xrightarrow{\alpha} \mathbb{R}^2 \xrightarrow{U} \mathscr{U}(\mathbb{C}^2)$
rph8-g15q_ EQ0043 (A10)	017	■ 1.000 ● N|+1	`f(\exp (x))=\exp (\mathcal{M}(x))`	$f(\exp (x))=\exp (\mathcal{M}(x))$	$f\big(\exp(x)\big) = \exp(\mathcal{M}(x))$
rph8-g15q_ EQ0044 (A11)	017	■ 1.000 ● K|+0	`\gamma(t)=\exp (t x)`	$\gamma(t)=\exp (t x)$	$\gamma(t) = \exp(tx)$
rph8-g15q_ EQ0045 (A12)	017	■ 1.000 ● N|+0	`(f \circ \gamma)(t)=\exp (t y)`	$(f \circ \gamma)(t)=\exp (t y)$	$(f \circ \gamma)(t) = \exp(ty)$
rph8-g15q_ EQ0047 (A14)	017	■ 1.000 ● K|+0	`f(\exp (t x))=\exp (t \mathcal{M}(x))`	$f(\exp (t x))=\exp (t \mathcal{M}(x))$	$f(\exp(tx)) = \exp(t\mathcal{M}(x))$
rph8-g15q_ EQ0048 (A15)	018	■ 1.000 ● K|+0	`f(x)=\exp (\mathcal{M}(x))`	$f(x)=\exp (\mathcal{M}(x))$	$f(x) = \exp(\mathcal{M}(x))$
rph8-g15q_ EQ0050 (B1)	018	■ 1.000 ● K|+0	`c\left(x_{1}, x_{2}\right)=c\left(x_{2}, x_{1}\right)=c \left(-x_{1},-x_{2}\right) .`	$c\left(x_{1}, x_{2}\right)=c\left(x_{2}, x_{1}\right)=c \left(-x_{1},-x_{2}\right) .$	$c(x_1, x_2) = c(x_2, x_1) = c(-x_1, -x_2).$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
rph8-g15q_ EQ0051 (B2)	018	■ 1.000 ● N|+0	`c(x)= \begin{cases}-1, & \text { if } y(x)<0, \\ 0, & \text { if } y(x)=0, \\ +1, & \text { if } y(x)>0, \end{cases}`	$c(x)= \begin{cases}-1, & \text { if } y(x)<0, \\ 0, & \text { if } y(x)=0, \\ +1, & \text { if } y(x)>0, \end{cases}$	$c(x)= \begin{cases}-1, & \text { if } y(x)<0, \\ 0, & \text { if } y(x)=0, \\ +1, & \text { if } y(x)>0, \end{cases}$
rph8-g15q_ EQ0052 (B3)	018	■ 1.000 ● N|+0	`y(x)=\operatorname{Tr}[\rho(x) O],`	$y(x)=\operatorname{Tr}[\rho(x) O],$	$y(x)=\operatorname{Tr}[\rho(x) O],$
rph8-g15q_ EQ0053 (B4)	018	■ 1.000 ● N|+0	`y\left(x_{1}, x_{2}\right)=y\left(x_{2}, x_{1}\right)=y \left(-x_{1},-x_{2}\right)`	$y\left(x_{1}, x_{2}\right)=y\left(x_{2}, x_{1}\right)=y \left(-x_{1},-x_{2}\right)$	$y(x_1, x_2)=y(x_2, x_1)=y(-x_1,-x_2)$
rph8-g15q_ EQ0055 (B6)	018	■ 1.000 ● K|+0	`O=X \otimes X,`	$O=X \otimes X,$	$O=X \otimes X,$
rph8-g15q_ EQ0056 (B7)	018	■ 1.000 ● K|+0	`y\left(x_{1}, x_{2}\right)=\cos \left(x_{1}\right) \sin \left(x_{2}\right)+2 \sqrt{p(1-p)} \sin \left(x_{1} \right) \sin \left(x_{2}\right) .`	$y\left(x_{1}, x_{2}\right)=\cos \left(x_{1}\right) \sin \left(x_{2}\right)+2 \sqrt{p(1-p)} \sin \left(x_{1} \right) \sin \left(x_{2}\right) .$	$y(x_1, x_2)=\cos(x_1)\sin(x_2)+2\sqrt{p(1-p)}\sin(x_1)\sin(x_2).$
rph8-g15q_ EQ0057 (B8)	018	■ 1.000 ● N|-1	`L_{1}=\left[\begin{array}{llll} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \end{array}\right],`	$L_{1}=\left[\begin{array}{llll} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \end{array}\right],$	$L_{1}=\left[\begin{array}{llll} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \end{array}\right],$
rph8-g15q_ EQ0058 (C3)	019	■ 1.000 ● W|+2	`\left[\begin{array}{llll} \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ a_{11} & a_{12} & a_{13} & a_{14} \end{array}\right]=-\left[\begin{array}{llll} a_{14} & a_{13} & a_{12} & a_{11} \\ a_{24} & a_{23} & a_{22} & \overline{a_{12}} \\ a_{34} & a_{33} & \overline{a_{23}} & \overline{a_{13}} \\ a_{44} & \overline{a_{34}} & \overline{a_{24}} & \overline{a_{14}} \end{array}\right] .`	$\left[\begin{array}{llll} \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ a_{11} & a_{12} & a_{13} & a_{14} \end{array}\right]=-\left[\begin{array}{llll} a_{14} & a_{13} & a_{12} & a_{11} \\ a_{24} & a_{23} & a_{22} & \overline{a_{12}} \\ a_{34} & a_{33} & \overline{a_{23}} & \overline{a_{13}} \\ a_{44} & \overline{a_{34}} & \overline{a_{24}} & \overline{a_{14}} \end{array}\right] .$	$\left[\begin{array}{llll} \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ a_{11} & a_{12} & a_{13} & a_{14} \end{array}\right]=-\left[\begin{array}{llll} a_{14} & a_{13} & a_{12} & a_{11} \\ a_{24} & a_{23} & a_{22} & \overline{a_{12}} \\ a_{34} & a_{33} & \overline{a_{23}} & \overline{a_{13}} \\ a_{44} & \overline{a_{34}} & \overline{a_{24}} & \overline{a_{14}} \end{array}\right] .$
rph8-g15q_ EQ0059 (B10)	019	■ 1.000 ● W|-2	`L_{1}=\left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & -\overline{a_{13}} \\ \overline{a_{13}} & -a_{23} & -a_{22} & -\overline{a_{12}} \\ -a_{14} & -a_{13} & -a_{12} & -a_{11} \end{array}\right],`	$L_{1}=\left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & -\overline{a_{13}} \\ \overline{a_{13}} & -a_{23} & -a_{22} & -\overline{a_{12}} \\ -a_{14} & -a_{13} & -a_{12} & -a_{11} \end{array}\right],$	$L_{1}=\left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & -\overline{a_{13}} \\ \overline{a_{13}} & -a_{23} & -a_{22} & -\overline{a_{12}} \\ -a_{14} & -a_{13} & -a_{12} & -a_{11} \end{array}\right],$
rph8-g15q_ EQ0060 (B11)	019	■ 1.000 ● N|+0	`a_{12}, a_{13} \in \mathbb{C}, \quad a_{14}, a_{23} \in i \mathbb{R}, \quad a_{11}, a_{22} \in \mathbb{R} .`	$a_{12}, a_{13} \in \mathbb{C}, \quad a_{14}, a_{23} \in i \mathbb{R}, \quad a_{11}, a_{22} \in \mathbb{R} .$	$a_{12}, a_{13} \in \mathbb{C}, \quad a_{14}, a_{23} \in i \mathbb{R}, \quad a_{11}, a_{22} \in \mathbb{R} .$
rph8-g15q_ EQ0062 (C1)	019	■ 1.000 ● N|+0	`d_{\mathcal{X}}\left(x_{1}, x_{2}\right)=\sqrt{ \left(x_{1}-x_{2}\right)^{T}\left(V^{T} V\right) \left(x_{1}-x_{2}\right)},`	$d_{\mathcal{X}}\left(x_{1}, x_{2}\right)=\sqrt{ \left(x_{1}-x_{2}\right)^{T}\left(V^{T} V\right) \left(x_{1}-x_{2}\right)},$	$d_{\mathcal{X}}(x_1, x_2)=\sqrt{(x_1-x_2)^T(V^TV)(x_1-x_2)},$
rph8-g15q_ EQ0063 (C2)	019	■ 1.000 ● K|+0	`d_{A}\left(x_{1}, x_{2}\right):=\sqrt{\left(x_{1}-x_{2} \right)^{T} A\left(x_{1}-x_{2}\right)} .`	$d_{A}\left(x_{1}, x_{2}\right):=\sqrt{\left(x_{1}-x_{2} \right)^{T} A\left(x_{1}-x_{2}\right)} .$	$d_A(x_1, x_2):=\sqrt{(x_1-x_2)^TA(x_1-x_2)}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

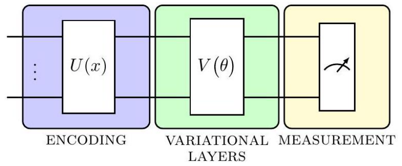
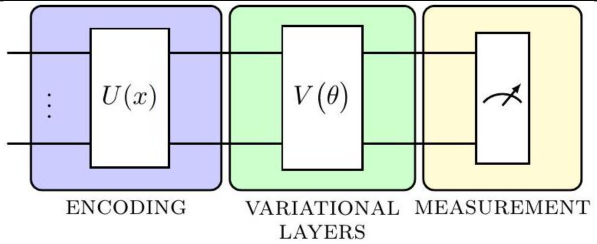
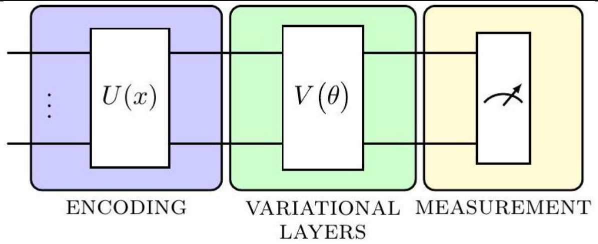
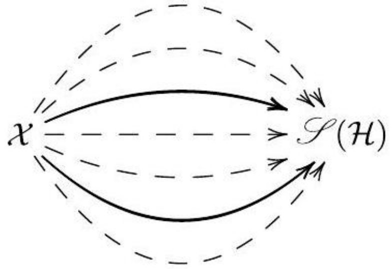
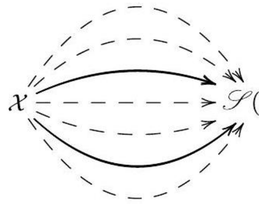
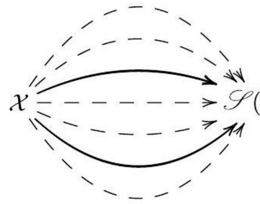
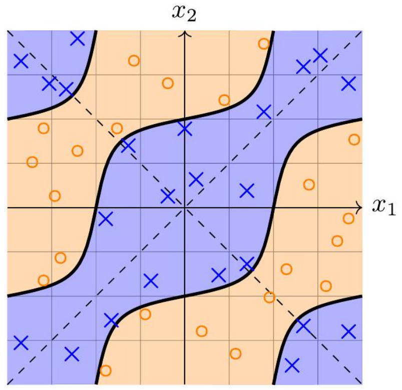
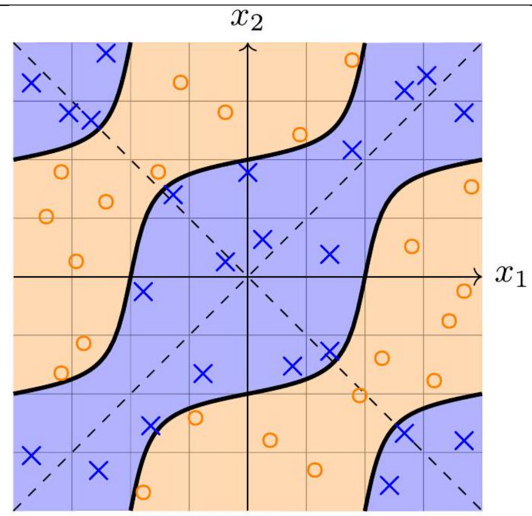
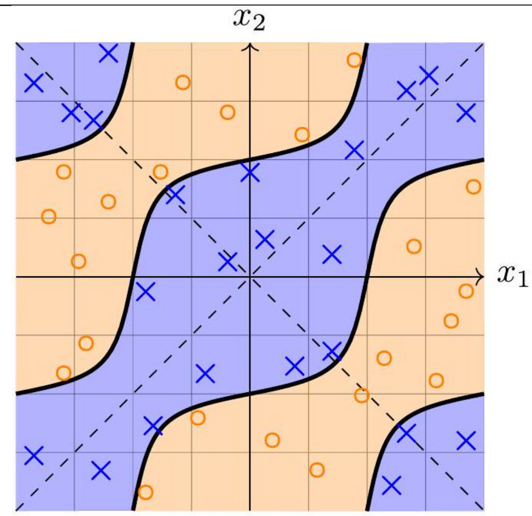
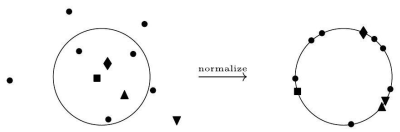
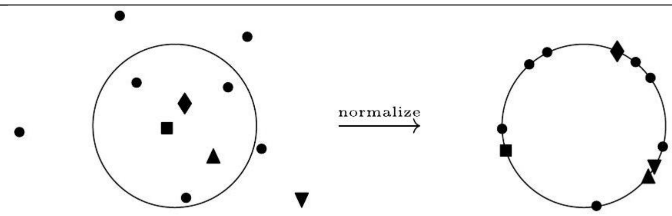
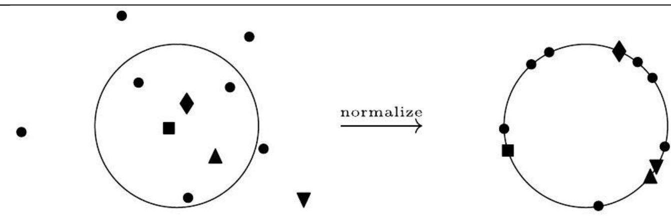
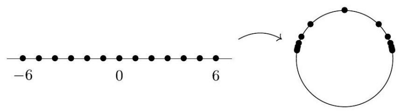
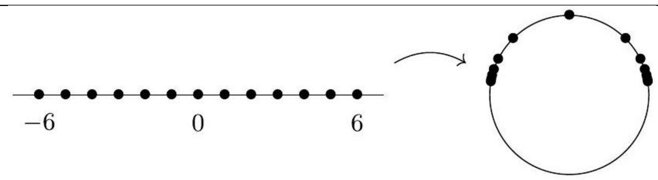
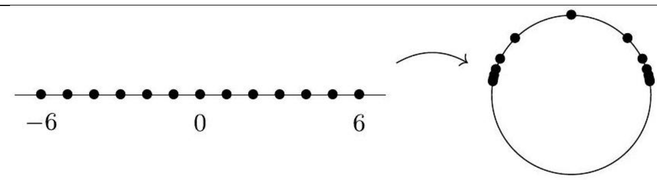
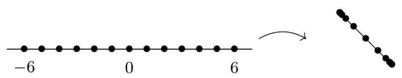
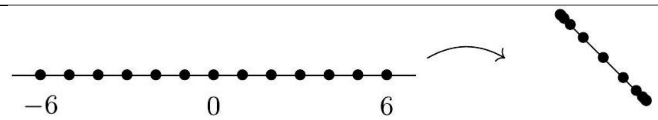
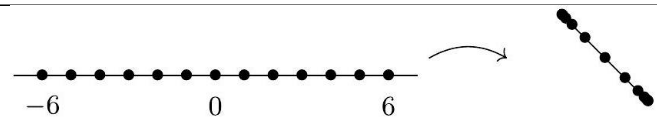
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

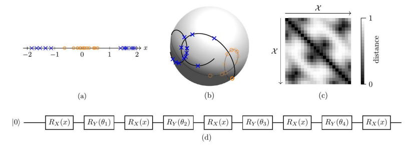
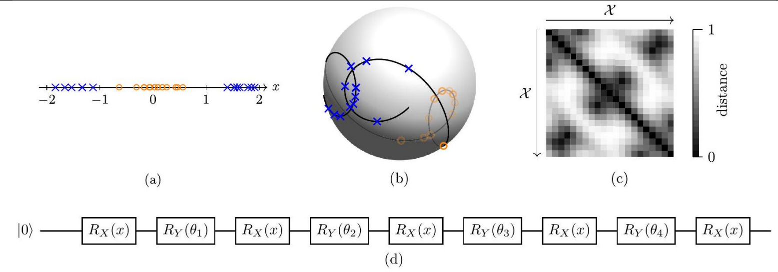
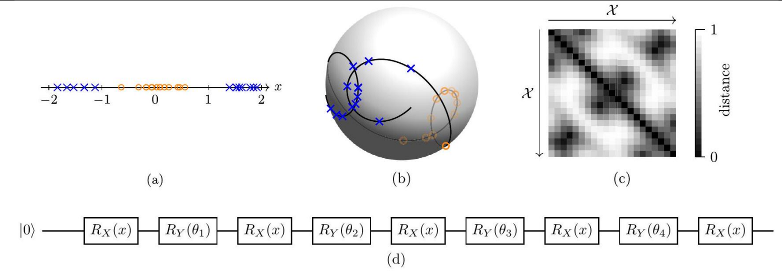
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
rph8-g15q_ EQ0066 (D2)	019	■ 1.000 ● W|-1	`\begin{aligned} d_{\mathcal{X}}\left(x_{1}, x_{2}\right) & =d_{\mathcal{Y}}\left(f\left(x_{1}\right), f\left(x_{2}\right)\right) \\ & \geqslant d_{\mathcal{Y}}\left(f \left(x_{1}\right), y\right)+d_{\mathcal{Y}}\left(y, f \left(x_{2}\right)\right) \end{aligned}`	$\begin{aligned} d_{\mathcal{X}}\left(x_{1}, x_{2}\right) & =d_{\mathcal{Y}}\left(f\left(x_{1}\right), f\left(x_{2}\right)\right) \\ & \geqslant d_{\mathcal{Y}}\left(f \left(x_{1}\right), y\right)+d_{\mathcal{Y}}\left(y, f \left(x_{2}\right)\right) \end{aligned}$	$\begin{aligned} d_{\mathcal{X}}(x_1, x_2) & = d_{\mathcal{Y}}(f(x_1), f(x_2)) \\ & \geqslant d_{\mathcal{Y}}(f(x_1), y) + d_{\mathcal{Y}}(y, f(x_2)) \end{aligned}$
rph8-g15q_ EQ0068 (D4)	019	■ 1.000 ● K|+0	`0=d_{\mathcal{X}}\left(x_{1}, x_{2}\right)=d_{ \mathcal{Y}}\left(f\left(x_{1}\right), f\left(x_{2} \right)\right),`	$0=d_{\mathcal{X}}\left(x_{1}, x_{2}\right)=d_{\mathcal{Y}}\left(f\left(x_{1}\right), f\left(x_{2}\right)\right),$	$0 = d_{\mathcal{X}}(x_1, x_2) = d_{\mathcal{Y}}(f(x_1), f(x_2)),$
rph8-g15q_ EQ0069 (D5)	020	■ 1.000 ● N|+0	`c(x)= \begin{cases}+1, & \text { if } x \in A, \\ -1, & \text { if } x \in B .\end{cases}`	$c(x)= \begin{cases}+1, & \text { if } x \in A, \\ -1, & \text { if } x \in B .\end{cases}$	$c(x) = \begin{cases} +1, & \text{if } x \in A, \\ -1, & \text{if } x \in B. \end{cases}$
rph8-g15q_ EQ0070 (D6)	020	■ 1.000 ● N|-1	`\rho_{A}:=\frac{1}{\# A} \sum_{a \in A} \rho(a) \quad \text { and } \quad \rho_{B}:=\frac{1}{\# B} \sum_{b \in B} \rho(b),`	$\rho_{A}:=\frac{1}{\# A} \sum_{a \in A} \rho(a) \quad \text { and } \quad \rho_{B}:=\frac{1}{\# B} \sum_{b \in B} \rho(b),$	$\rho_A := \frac{1}{\#A} \sum_{a \in A} \rho(a) \quad \text{and} \quad \rho_B := \frac{1}{\#B} \sum_{b \in B} \rho(b),$
rph8-g15q_ EQ0071 (D7)	020	■ 1.000 ● K|+0	`C\left(\rho_{A}, \rho_{B}\right)=1-\frac{1}{2} d_{ \mathrm{HS}}\left(\rho_{A}, \rho_{B}\right) .`	$C\left(\rho_{A}, \rho_{B}\right)=1-\frac{1}{2} d_{\mathrm{HS}}\left(\rho_{A}, \rho_{B}\right) .$	$C(\rho_A, \rho_B) = 1 - \frac{1}{2} d_{\mathrm{HS}}(\rho_A, \rho_B).$
rph8-g15q_ EQ0072 (D8)	020	■ 1.000 ● N|+0	`U_{\theta}(x)=\left(\prod_{n=1}^{4} R_{X}(x) R_{Y}\left( \theta_{n}\right)\right) R_{X}(x),`	$U_{\theta}(x)=\left(\prod_{n=1}^{4} R_{X}(x) R_{Y}\left(\theta_{n}\right)\right) R_{X}(x),$	$U_\theta(x) = \left(\prod_{n=1}^{4} R_X(x) R_Y(\theta_n) \right) R_X(x),$
rph8-g15q_ EQ0073 (D9)	020	■ 1.000 ● N|+0	`\prod_{n=1}^{N} A_{n}=A_{N} \cdots A_{1} .`	$\prod_{n=1}^{N} A_{n}=A_{N} \cdots A_{1} .$	$\prod_{n=1}^{N} A_n = A_N \cdots A_1.$
rph8-g15q_ EQ0074 (D10)	020	■ 1.000 ● K|+0	`\rho_{\theta}(x)=|x\rangle\langle x|,`	$\rho_{\theta}(x)=\lvert x\rangle\langle x\rvert,$	$\rho_\theta(x) = \lvert x\rangle\langle x\rvert,$
rph8-g15q_ EQ0075 (D11)	020	■ 1.000 ● K|+0	`|x\rangle=U_{\theta}(x)|0\rangle .`	$\lvert x\rangle=U_{\theta}(x)\lvert 0\rangle .$	$\lvert x\rangle = U_\theta(x)\lvert 0\rangle.$
rph8-g15q_ EQ0076 (D12)	020	■ 1.000 ● K|+0	`O:=\rho_{A}-\rho_{B},`	$O:=\rho_{A}-\rho_{B},$	$O := \rho_A - \rho_B,$
rph8-g15q_ EQ0077 (D13)	020	■ 1.000 ● N|+1	`y(x)=\operatorname{Tr}[\rho(x) O] .`	$y(x)=\operatorname{Tr}[\rho(x) O] .$	$y(x) = \operatorname{Tr}\big[\rho(x)O\big].$
rph8-g15q_ EQ0078 (D14)	020	■ 1.000 ● K|+0	`\operatorname{Tr}\left[\rho_{A}-\rho_{B}\right]= \operatorname{Tr}\left[\rho_{A}\right]-\operatorname{Tr} \left[\rho_{B}\right]=1-1=0,`	$\operatorname{Tr}\left[\rho_{A}-\rho_{B}\right]=\operatorname{Tr}\left[\rho_{A}\right]-\operatorname{Tr}\left[\rho_{B}\right]=1-1=0,$	$\operatorname{Tr}[\rho_A - \rho_B] = \operatorname{Tr}[\rho_A] - \operatorname{Tr}[\rho_B] = 1 - 1 = 0,$
rph8-g15q_ EQ0079 (D15)	020	■ 1.000 ● K|+0	`\operatorname{sgn}(y(x))=c(x)`	$\operatorname{sgn}(y(x))=c(x)$	$\operatorname{sgn}\big(y(x)\big) = c(x)$
rph8-g15q_ EQ0080 (E2)	021	■ 1.000 ● W|+0	`\overbrace{\mathbb{C}^{2} \otimes \cdots \otimes \mathbb{C}^{2}}^{n \text { times }} \cong \mathbb{C}^{2^{n}},`	$\overbrace{\mathbb{C}^{2} \otimes \cdots \otimes \mathbb{C}^{2}}^{n \text { times }} \cong \mathbb{C}^{2^{n}},$	$\overbrace{\mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2}^{n \text{ times}} \cong \mathbb{C}^{2^n},$
rph8-g15q_ EQ0081 (E3)	021	■ 1.000 ● K|+0	`\left|x_{0} x_{1} \cdots x_{n-1}\right\rangle=\left|x_{0} \right\rangle \otimes\left|x_{1}\right\rangle \otimes \cdots \otimes\left|x_{n-1}\right\rangle .`	$\left\lvert x_{0} x_{1} \cdots x_{n-1}\right\rangle=\left\lvert x_{0}\right\rangle \otimes\left\lvert x_{1}\right\rangle \otimes \cdots \otimes\left\lvert x_{n-1}\right\rangle .$	$\lvert x_0 x_1 \cdots x_{n-1}\rangle = \lvert x_0\rangle \otimes \lvert x_1\rangle \otimes \cdots \otimes \lvert x_{n-1}\rangle.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Image regions — rendered against scan

Two comparable cells per row. **Rendered**: the region's own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan**: the `tex.zip` image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix's confidence, as the equation rows do. Setting a region's LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with `pdfdrill vision`; verify any LLM result against the real ink with inkdrill.

Identifier	Page	Class	Source	Author source	Rendered	Scan
rph8-g15q_DIA_0001	003	— *(dup)*	2ea50605-a8ad-43af-b377-03a13f1f568b-03_316_769_242_213.jpg			
rph8-g15q_DIA_0002	004	— *(dup)*	2ea50605-a8ad-43af-b377-03a13f1f568b-04_264_380_242_409.jpg			
rph8-g15q_DIA_0003	004	— *(dup)*	2ea50605-a8ad-43af-b377-03a13f1f568b-04_664_677_242_1174.jpg			
rph8-g15q_DIA_0004	007	— *(dup)*	2ea50605-a8ad-43af-b377-03a13f1f568b-07_275_866_240_1079.jpg			
rph8-g15q_DIA_0005	007	— *(dup)*	2ea50605-a8ad-43af-b377-03a13f1f568b-07_229_844_2101_1090.jpg			
rph8-g15q_DIA_0006	008	— *(dup)*	2ea50605-a8ad-43af-b377-03a13f1f568b-08_149_846_240_178.jpg			

rph8-g15q_DIA_0007	008	— *(dup)*	2ea50605-a8ad-43af-b377-03a13f1f568b-08_178_866_2228_167.jpg	$(X,d_X) \longmapsto K_\bullet(X,d_X) \longmapsto H_\bullet(X,d_X) \longmapsto \mathrm{dgm}_\bullet(X,d_X)$ finite metric space, filtered simplicial complex, persistence vector space, persistence diagram	$(X,d_X) \longmapsto K_\bullet(X,d_X) \longmapsto H_\bullet(X,d_X) \longmapsto \mathrm{dgm}_\bullet(X,d_X)$ finite metric space, filtered simplicial complex, persistence vector space, persistence diagram	$(X,d_X) \longmapsto K_\bullet(X,d_X) \longmapsto H_\bullet(X,d_X) \longmapsto \mathrm{dgm}_\bullet(X,d_X)$ finite metric space, filtered simplicial complex, persistence vector space, persistence diagram
rph8-g15q_DIA_0008	011	— *(dup)*	2ea50605-a8ad-43af-b377-03a13f1f568b-11_618_1736_237_202.jpg			
rph8-g15q_DIA_0009	013	— *(dup)*	2ea50605-a8ad-43af-b377-03a13f1f568b-13_352_732_237_231.jpg	$G\text{-}\mathbf{Set} \xrightarrow{\text{forget action}} \mathbf{Set}$ data domain $(\mathcal{X},\alpha) \xmapsto{\text{forget } \alpha} \mathcal{X}$ state space $(\mathscr{S}(\mathcal{H}),\mathrm{Ad}_V) \xmapsto{\text{forget } \mathrm{Ad}_V} \mathscr{S}(\mathcal{H})$ arbitrary G-set $(\mathcal{Y},\beta) \xmapsto{\text{forget } \beta} \mathcal{Y}$	$G\text{-}\mathbf{Set} \xrightarrow{\text{forget action}} \mathbf{Set}$ data domain $(\mathcal{X},\alpha) \xmapsto{\text{forget } \alpha} \mathcal{X}$ state space $(\mathscr{S}(\mathcal{H}),\mathrm{Ad}_V) \xmapsto{\text{forget } \mathrm{Ad}_V} \mathscr{S}(\mathcal{H})$ arbitrary G-set $(\mathcal{Y},\beta) \xmapsto{\text{forget } \beta} \mathcal{Y}$	$G\text{-}\mathbf{Set} \xrightarrow{\text{forget action}} \mathbf{Set}$ data domain $(\mathcal{X},\alpha) \xmapsto{\text{forget } \alpha} \mathcal{X}$ state space $(\mathscr{S}(\mathcal{H}),\mathrm{Ad}_V) \xmapsto{\text{forget } \mathrm{Ad}_V} \mathscr{S}(\mathcal{H})$ arbitrary G-set $(\mathcal{Y},\beta) \xmapsto{\text{forget } \beta} \mathcal{Y}$
rph8-g15q_DIA_0010	014	— *(dup)*	2ea50605-a8ad-43af-b377-03a13f1f568b-14_224_716_2004_240.jpg	$\mathbf{C} \xrightarrow{F} \mathbf{Set}$ $\quad \begin{array}{ccc} \mathcal{X}' & \overset{\rho'}{\dashrightarrow} & \mathscr{S}(\mathcal{H})' \\ \downarrow & & \downarrow \\ \mathcal{X} & \xrightarrow[\rho]{} & \mathscr{S}(\mathcal{H}) \end{array}$	$\mathbf{C} \xrightarrow{F} \mathbf{Set}$ $\quad \begin{array}{ccc} \mathcal{X}' & \overset{\rho'}{\dashrightarrow} & \mathscr{S}(\mathcal{H})' \\ \downarrow & & \downarrow \\ \mathcal{X} & \xrightarrow[\rho]{} & \mathscr{S}(\mathcal{H}) \end{array}$	$\mathbf{C} \xrightarrow{F} \mathbf{Set}$ $\quad \begin{array}{ccc} \mathcal{X}' & \overset{\rho'}{\dashrightarrow} & \mathscr{S}(\mathcal{H})' \\ \downarrow & & \downarrow \\ \mathcal{X} & \xrightarrow[\rho]{} & \mathscr{S}(\mathcal{H}) \end{array}$
rph8-g15q_DIA_0011	014	— *(dup)*	2ea50605-a8ad-43af-b377-03a13f1f568b-14_218_782_763_1121.jpg	$G\text{-}\mathbf{Set} \xrightarrow{F} \mathbf{Set}$ $\quad \begin{array}{ccc} (\mathcal{X},\alpha) & \overset{\rho}{\dashrightarrow} & (\mathscr{S}(\mathcal{H}),\mathrm{Ad}_V) \\ \downarrow & & \downarrow \\ \mathcal{X} & \xrightarrow[\rho]{} & \mathscr{S}(\mathcal{H}) \end{array}$	$G\text{-}\mathbf{Set} \xrightarrow{F} \mathbf{Set}$ $\quad \begin{array}{ccc} (\mathcal{X},\alpha) & \overset{\rho}{\dashrightarrow} & (\mathscr{S}(\mathcal{H}),\mathrm{Ad}_V) \\ \downarrow & & \downarrow \\ \mathcal{X} & \xrightarrow[\rho]{} & \mathscr{S}(\mathcal{H}) \end{array}$	$G\text{-}\mathbf{Set} \xrightarrow{F} \mathbf{Set}$ $\quad \begin{array}{ccc} (\mathcal{X},\alpha) & \overset{\rho}{\dashrightarrow} & (\mathscr{S}(\mathcal{H}),\mathrm{Ad}_V) \\ \downarrow & & \downarrow \\ \mathcal{X} & \xrightarrow[\rho]{} & \mathscr{S}(\mathcal{H}) \end{array}$
rph8-g15q_DIA_0012	021	— *(dup)*	2ea50605-a8ad-43af-b377-03a13f1f568b-21_240_358_1248_420.jpg	$\begin{array}{ccc} F(\mathcal{X}) & \xrightarrow{F(f)} & F(\mathcal{Y}) \\ \downarrow{\scriptstyle \eta_{\mathcal{X}}} & & \downarrow{\scriptstyle \eta_{\mathcal{Y}}} \\ G(\mathcal{X}) & \xrightarrow[G(f)]{} & G(\mathcal{Y}) \end{array}$	$\begin{array}{ccc} F(\mathcal{X}) & \xrightarrow{F(f)} & F(\mathcal{Y}) \\ \downarrow{\scriptstyle \eta_{\mathcal{X}}} & & \downarrow{\scriptstyle \eta_{\mathcal{Y}}} \\ G(\mathcal{X}) & \xrightarrow[G(f)]{} & G(\mathcal{Y}) \end{array}$	$\begin{array}{ccc} F(\mathcal{X}) & \xrightarrow{F(f)} & F(\mathcal{Y}) \\ \downarrow{\scriptstyle \eta_{\mathcal{X}}} & & \downarrow{\scriptstyle \eta_{\mathcal{Y}}} \\ G(\mathcal{X}) & \xrightarrow[G(f)]{} & G(\mathcal{Y}) \end{array}$